# FINAL CHALLENGE

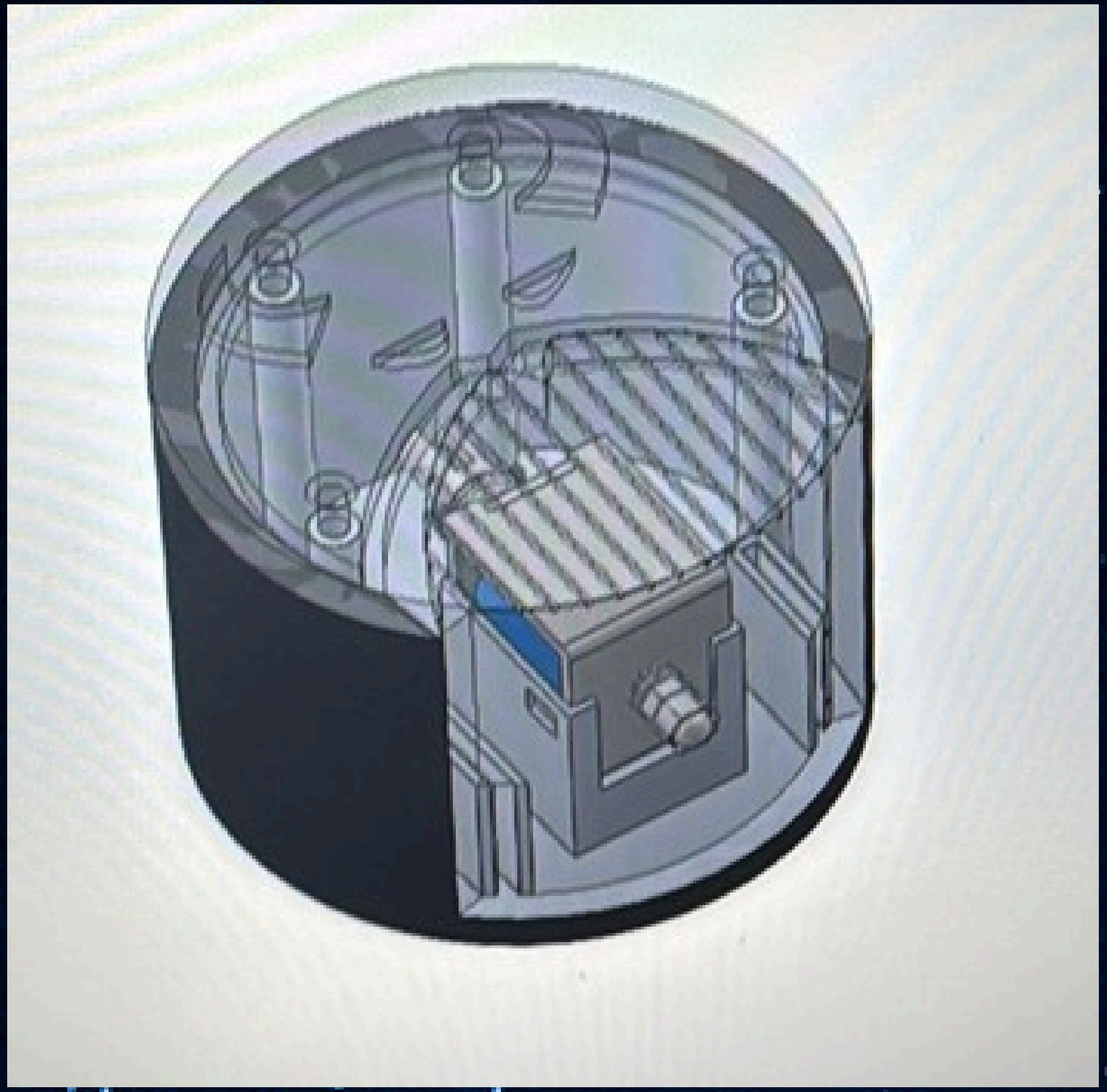
MONTSERRAT MARTÍNEZ GONZÁLEZ A00839323
REBECA LARA CORTÉS A01659574
SOFÍA JAQUELINE BARAJAS PÉREZ A01571651
JOSÉ PABLO RODRÍGUEZ PINAL A01752350

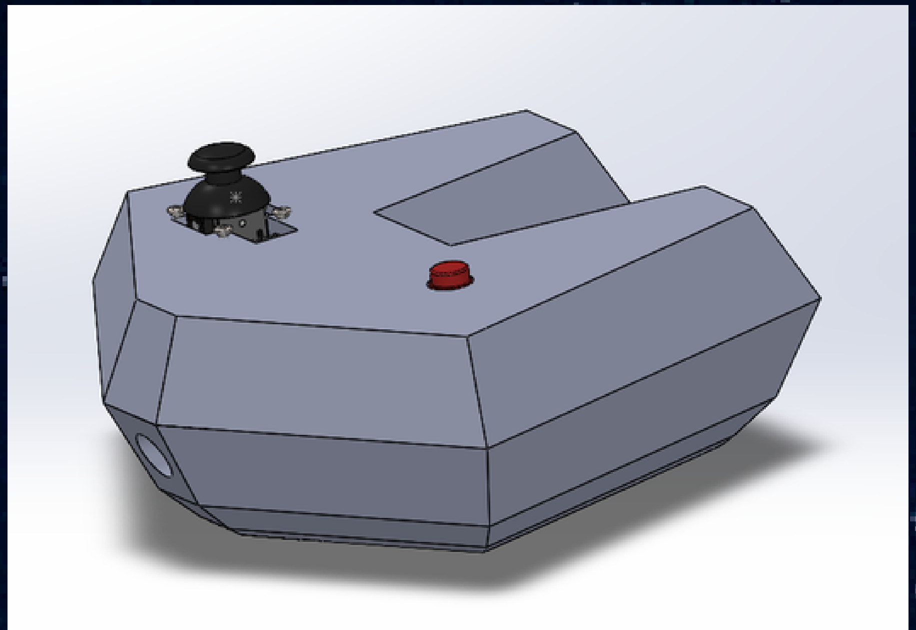
# GOALKEEPER

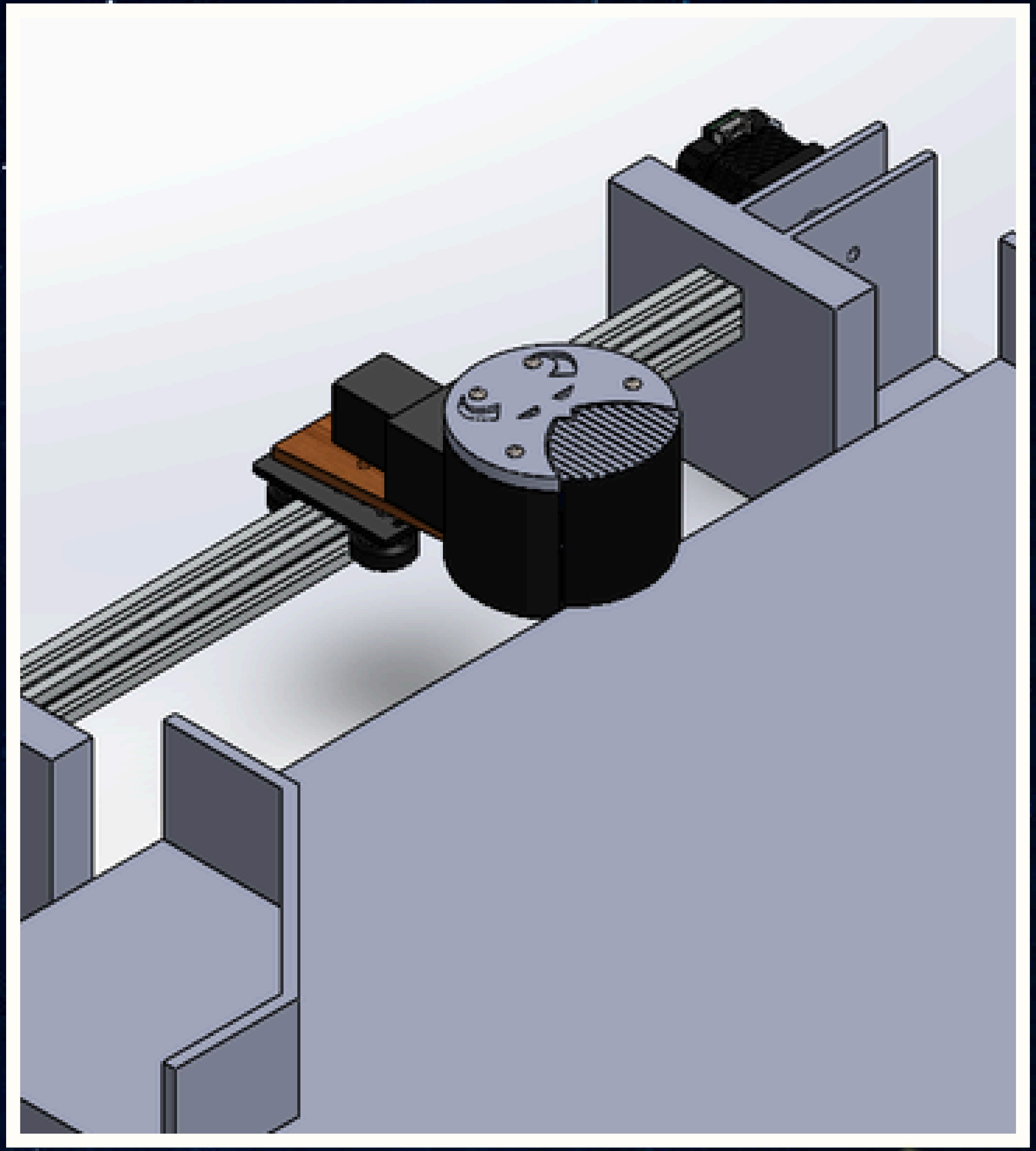
The objective of this project was to design and build an interactive robotic goalkeeper capable of moving laterally along the goal while simulating a kicking motion using a solenoid actuator.

The system combines an Arduino-based control system, a stepper motor with linear motion transmission, a custom electronic circuit, and 3D-printed mechanical components.
idunt pharetra leo. Cras facilisis leo non nibh congue volutpat.

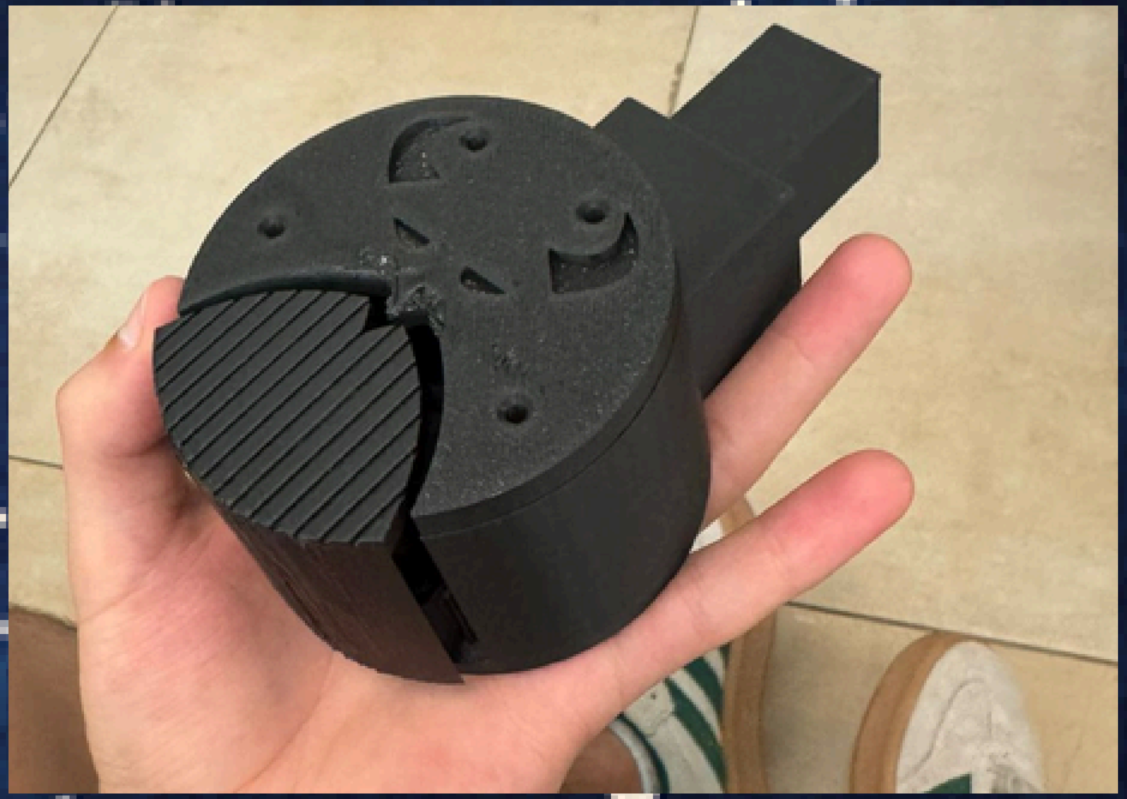
# CAD

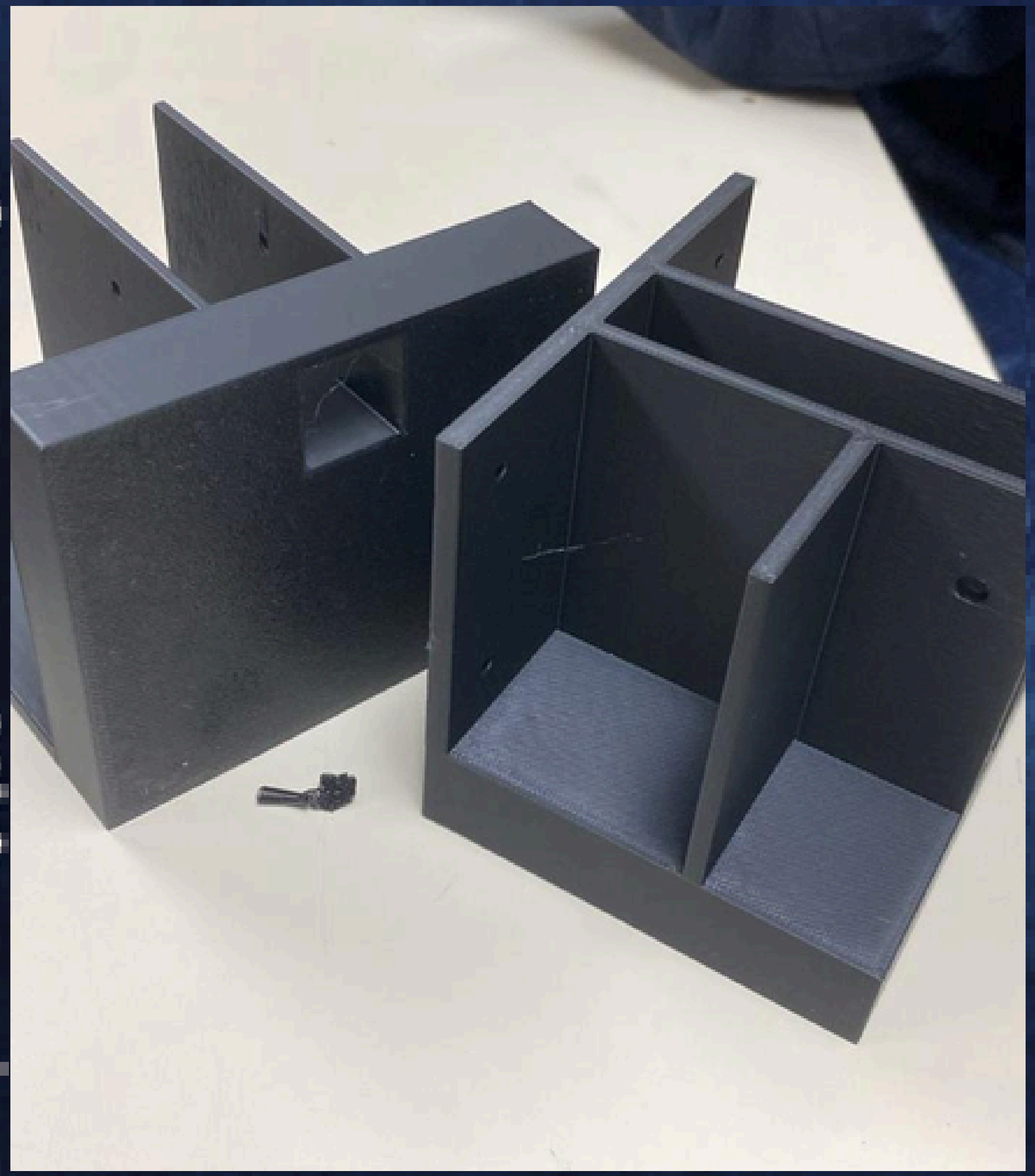




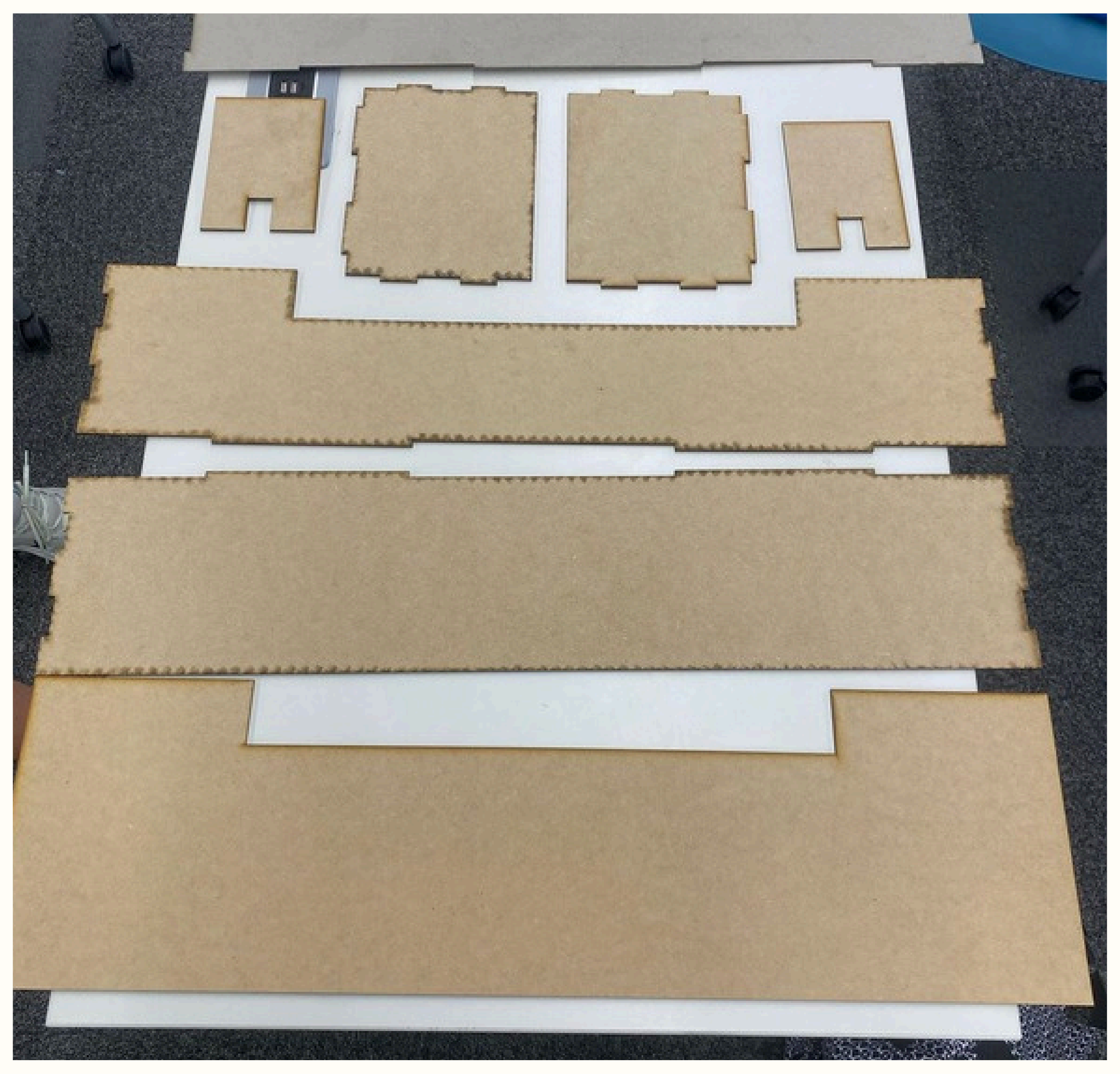


# MANUFACTURE

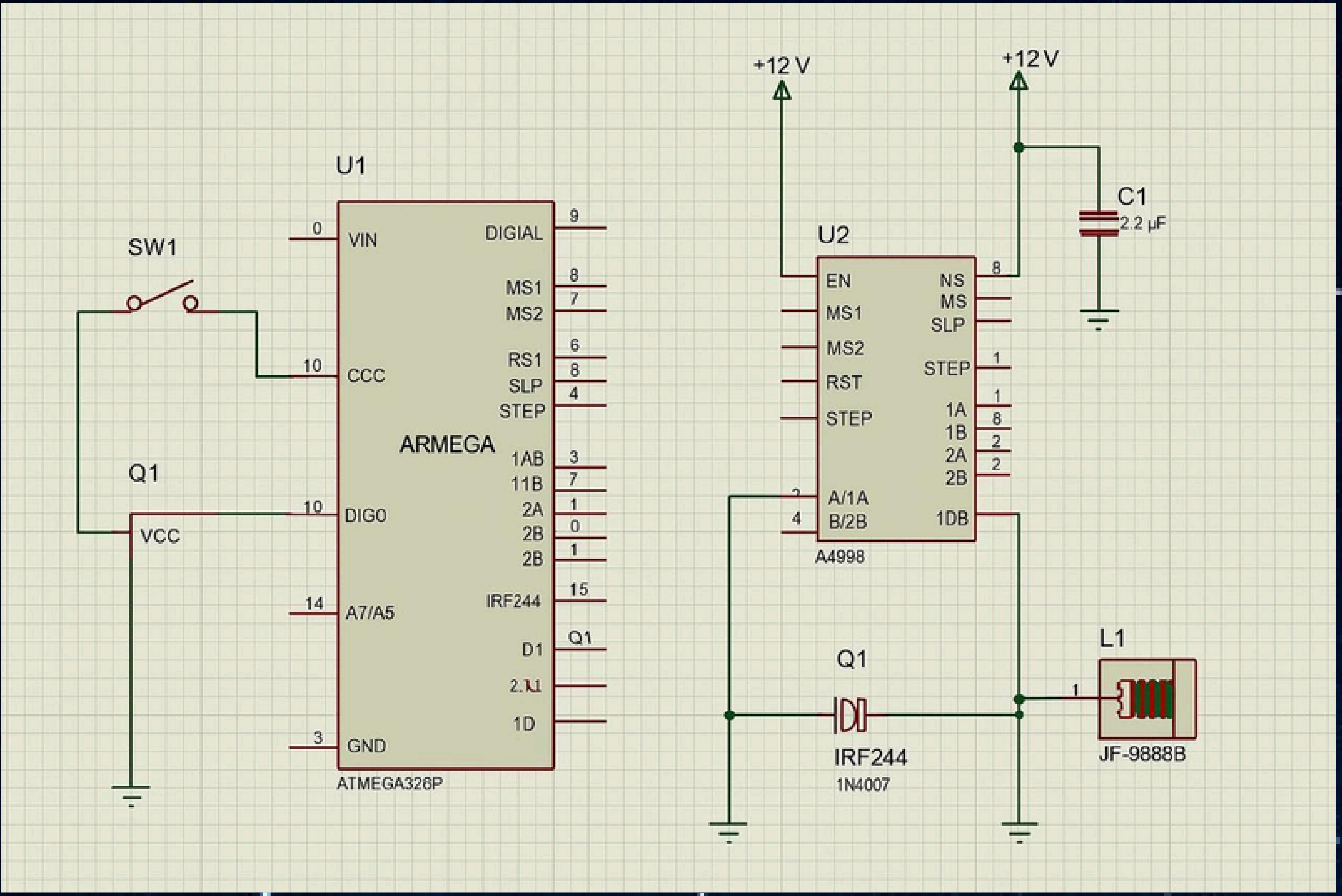


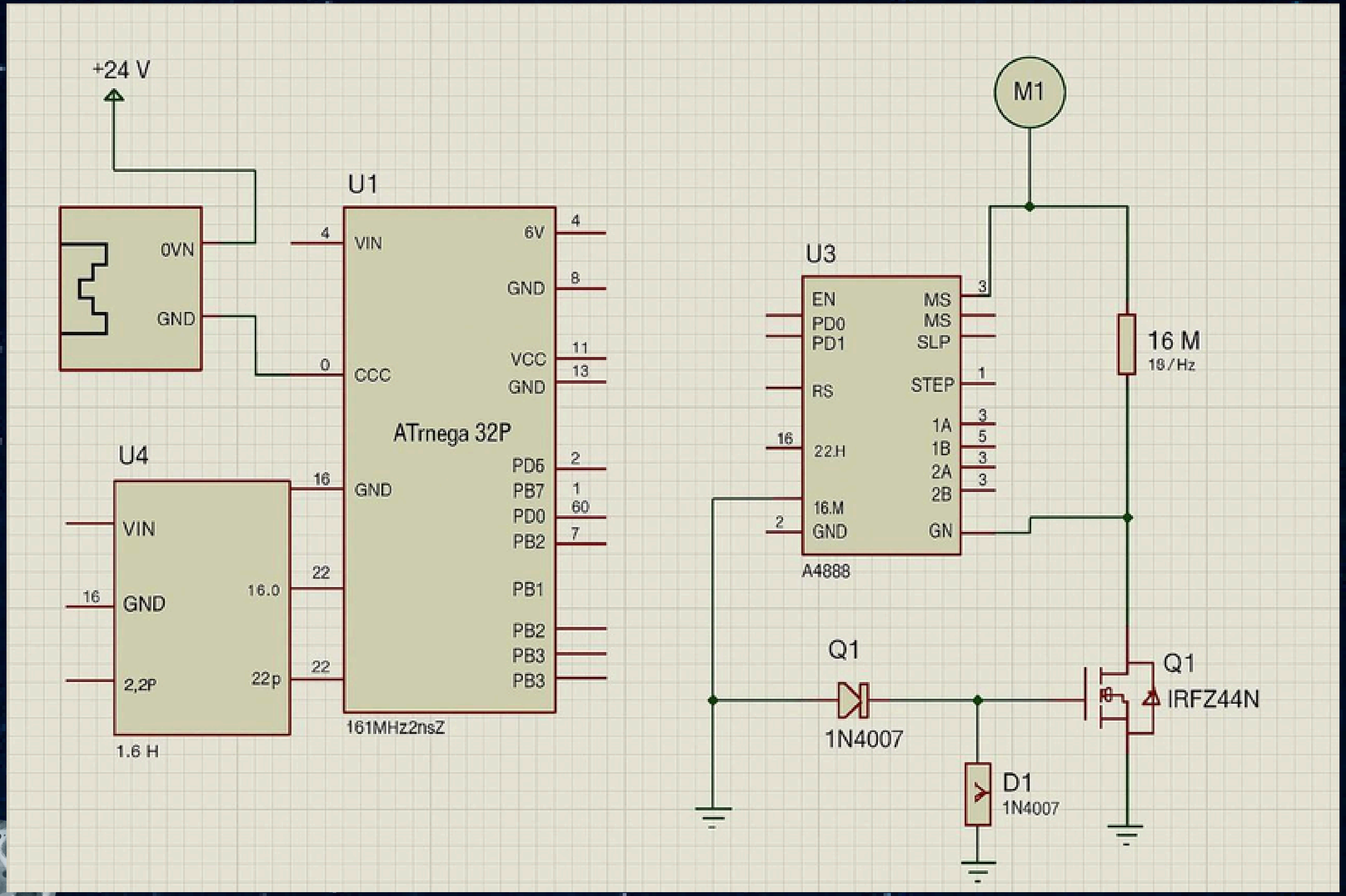


# MANUFACTURE

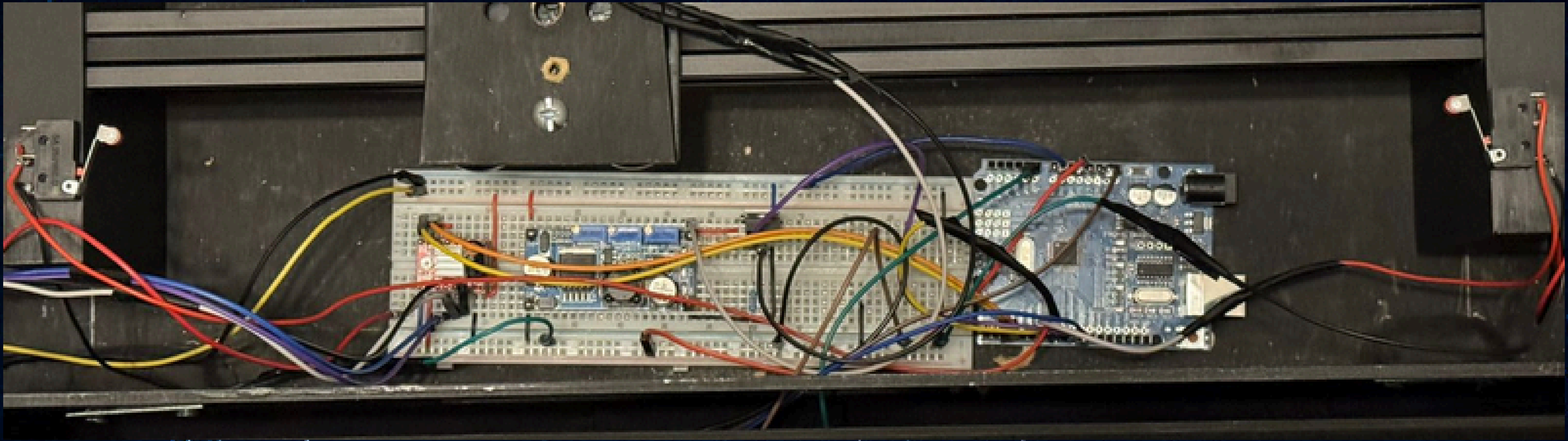


# ELECTRONICS



# ELECTRONICS



# CODE

GOALKEEPER_FINAL_CODE.ino

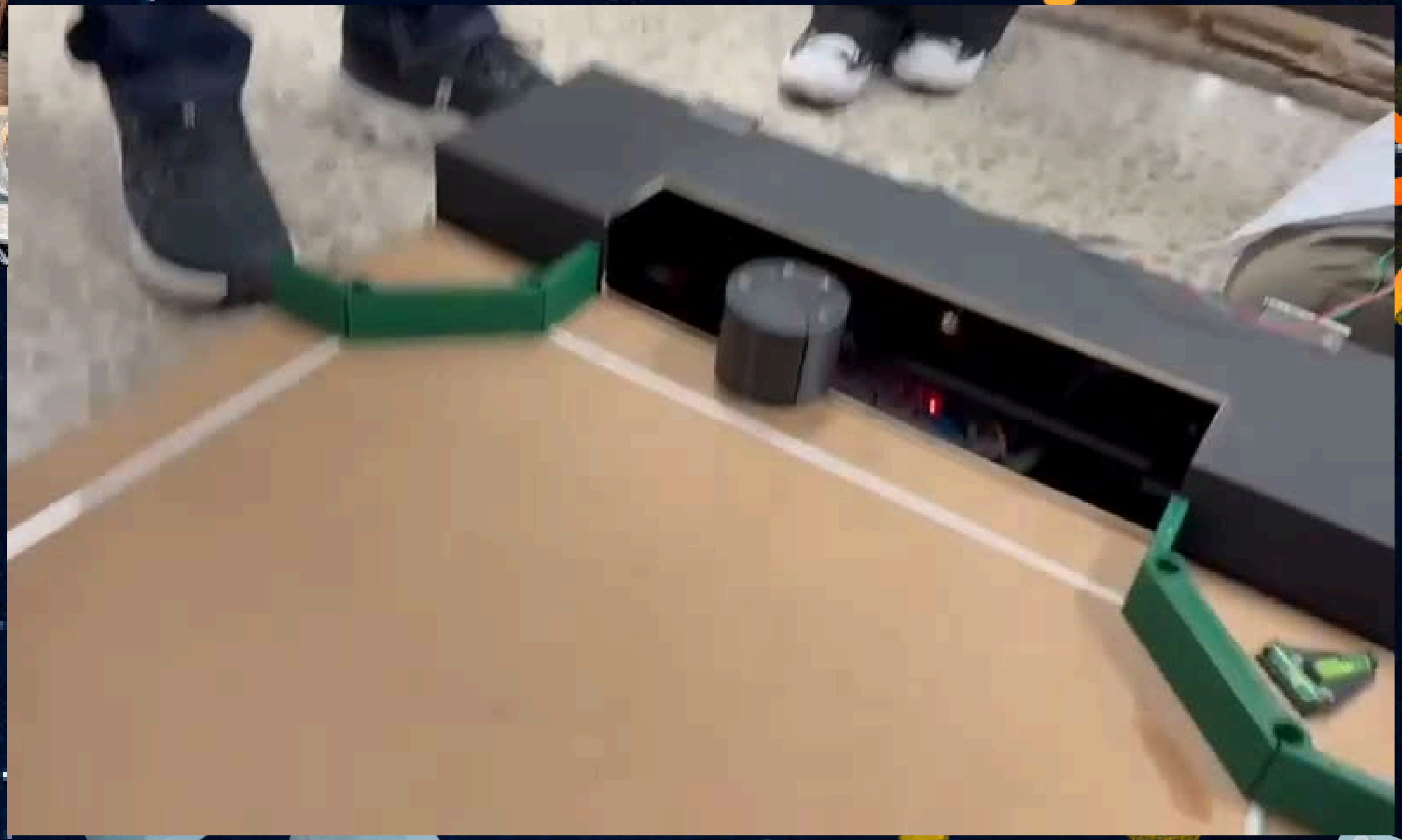
```
const int dirPin = 2;
const int stepPin = 3;

int xPin = A0;
int xVal;

int ButtonPin = 10;
int ButtonState = 0;

int MosfetPin = 9;
int LeftSwitchPin = 7;
int RightSwitchPin = 6;

int LeftSwitchState = 0;
int RightSwitchState = 0;
int dt = 50;

int stepsPerRevolution = 200;
int stepDelay = 2;


void setup() {
  pinMode(dirPin, OUTPUT);
  pinMode(stepPin, OUTPUT);
  pinMode(xPin, INPUT);

  pinMode(ButtonPin, INPUT_PULLUP);      // Botón cor
  pinMode(MosfetPin, OUTPUT);            // Salida pa

  pinMode(LeftSwitchPin, INPUT_PULLUP);
  pinMode(RightSwitchPin, INPUT_PULLUP);

  digitalWrite(MosfetPin, LOW);          // Asegura c
```

GOALKEEPER_FINAL_CODE.ino

```

  Serial.begin(9600);
}

void loop() {

  ButtonState = digitalRead(ButtonPin);

  //  Control del solenoide
  if (ButtonState == LOW) {              // Botón pre
    digitalWrite(MosfetPin, HIGH);       // Activa el
  } else {
    digitalWrite(MosfetPin, LOW);        // Desactiva
  }

  xVal = analogRead(xPin);
  LeftSwitchState = digitalRead(LeftSwitchPin);
  RightSwitchState = digitalRead(RightSwitchPin);

  // Depuración
  Serial.print("X value = ");
  Serial.print(xVal);
  Serial.print(" - Left Switch = ");
  Serial.print(LeftSwitchState);
  Serial.print(" - Right Switch = ");
  Serial.print(RightSwitchState);
  Serial.print(" - Button = ");
  Serial.println(ButtonState);

  //  Control del motor a pasos basado en el joystic
  if (RightSwitchState == HIGH && xVal < 450) {
    digitalWrite(dirPin, HIGH); // Derecha
    for (int i = 0; i < 100; i++) {
```

```
      digitalWrite(stepPin, HIGH);
      delayMicroseconds(1000);
      digitalWrite(stepPin, LOW);
      delayMicroseconds(1000);
    }
  }
  else if (LeftSwitchState == HIGH && xVal > 650) {
    digitalWrite(dirPin, LOW); // Izquierda
    for (int i = 0; i < 100; i++) {
      digitalWrite(stepPin, HIGH);
      delayMicroseconds(1000);
      digitalWrite(stepPin, LOW);
      delayMicroseconds(1000);
    }
  }
}
```

# FINAL MODEL

# THANK YOU